

The Effects of Diaspirin Cross-Linked Hemoglobin on Hemodynamics, Metabolic Acidosis, and Survival in Burned Rats

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Objective: Diaspirin cross-linked hemoglobin (DCLHb) is a vasoactive hemoglobin-based oxygen carrier or "blood substitute" that has been shown to improve base deficit in several experimental studies of hemorrhagic shock. Our objective was to determine if the addition of DCLHb to the resuscitation regimen would improve hemodynamic parameters, metabolic acidosis, and survival in our rat burn shock model compared with currently used resuscitation therapy.

Methods: This was a randomized, controlled, experimental rat study. Male Wistar rats, weighing 200 to 250 g, were surgically prepared for an acute study. After placement of indwelling catheters, baseline hemodynamic values (mean arterial pressure, cardiac output, systemic vascular resistance, stroke volume, and base excess) were obtained. Thirty-two rats were used in the study, and they were either subjected to a 30% scald burn (experimental group) or sham burned (control group). The experimental animals were immediately intravenously resuscitated and followed for 6 hours. The resuscitation was based on the Parkland formula (4 mL/kg for each 1% of total body surface area [TBSA] burn), with 50% of the calculated fluid amount to be administered at a constant rate during the first 8 hours after burn. The animals were resuscitated for 6 hours and received between 9.00 and 11.25 mL of fluid depending on weight. The experimental animals were randomly assigned to one of three treatment groups: group I, lactated Ringer's solution; group II, lactated Ringer's solution-human serum albumin; group III, lactated Ringer's solution-DCLHb. Group I (n = 8) received 4 mL/kg lactated Ringer's solution for each 1% of TBSA burn. Group II (n = 8) received 2 mL/kg lactated Ringer's solution and 2 mL/kg human serum albumin for each 1% of TBSA burn. Group III (n = 8) received 2 mL/kg lactated Ringer's solution and 2 mL/kg DCLHb for each 1% of TBSA burn. The sham group (n = 8) was not burned and was not resuscitated. Animals that survived up to 6 hours were killed.

Results: We found that mean arterial pressure, cardiac output, stroke volume, and base excess were all improved in the DCLHb-lactated Ringer's solution-treated animals compared with the other experimental treatment groups. The 6-hour mortality rates were zero of eight (lactated Ringer's solution-DCLHb group), zero of eight (sham group), three of eight (lactated Ringer's solution-human serum albumin group), and six of eight (lactated Ringer's solution only group).

Conclusion: Early resuscitation with DCLHb is superior to non-oxygen-carrying resuscitative fluids in improving hemodynamics and survival in this model of burn shock. DCLHb might improve general tissue perfusion in the acute postburn period, and it could be useful in the early management of patients with severe burns.

Key Words: Diaspirin cross-linked hemoglobin, DCLHb, Shock, Burn, Thermal injury, Blood substitutes, Resuscitation, Critical care, Rats, Hemoglobin-based oxygen carriers.

Severe burn injury in which more than 20% of the body surface is burned results in an acute period of shock and increased capillary permeability in which intravascular fluid shifts occur, resulting in edema formation.¹ Elevated levels of vasoactive hormones are believed to be responsible for the increased systemic vascular resistance and the decreased cardiac output and cardiac contractility that frequently accompany burn shock, coupled with decreased blood flow (perfusion) to many tissues and organs.² Patients usually develop a severe metabolic acidosis and base deficit that may be refractory to standard resuscitative fluids (i.e., lactated Ringer's solution or other crystalloids and colloids).

Diaspirin cross-linked hemoglobin (DCLHb; Baxter

Healthcare Corp., Round Lake, Ill) is a modified stroma-free hemoglobin that is stabilized by intramolecular cross-linking of the α -subunits.³ DCLHb is a nonantigenic, low-viscosity solution that has O₂ transport properties similar to whole blood and has been subjected to rigorous virus-inactivation processes. The half-life of DCLHb is 3 to 24 hours, depending on the species and dose.^{4,5} DCLHb was previously shown to be efficacious as a resuscitative fluid in hemorrhagic shock and is equal to blood and superior to lactated Ringer's solution in restoring mean arterial pressure, base excess, central venous O₂ saturation, and transcutaneous Po₂.⁶⁻⁹ Additionally, DCLHb increases mean arterial pressure at low doses through a mechanism that might involve the inhibition of nitric oxide.¹⁰ Because DCLHb carries O₂ and improves perfusion, we tested the hypothesis that DCLHb could be useful in the treatment of burn shock in combination with the use of standard resuscitative fluids and other supportive care.

MATERIALS AND METHODS

The research protocol was approved by the University of California San Diego Laboratory Animal Review Board for

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This study was supported, in part, by a grant from Baxter Healthcare Corp. and the Burn Institute of San Diego (grant 96-5898).

Abstract presentation at the 29th Annual Meeting of the American Burn Association, March 19-22, 1997, New York, New York.

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the care and use of animals. The study was conducted in compliance with the Animal Welfare Act and other Federal statutes and regulations relating to animals and experiments involving animals, and it adhered to the principles stated in the "Guide for the Care and Use of Laboratory Animals" (National Institutes of Health publication 86-23, 1985 edition).

Male Wistar rats weighing 200 to 250 g were used in this study. They were maintained on a 12-hour light cycle and provided with water ad libitum throughout the study. Animals were anesthetized intraperitoneally with pentobarbital (35–45 mg/kg). The dorsum of the rat was shaved and depilated using Nair (Carter Wallace Inc., New York, NY). The next morning, the animals were again anesthetized using pentobarbital intraperitoneally. Using aseptic technique, the internal jugular vein and femoral artery were catheterized with polyethylene catheters (PE-50, Clay Adams, Piscataway, NJ). For the measurement of cardiac output, a thermistor probe (Columbus Instruments, Columbus, Ohio) was placed in the right carotid artery. The catheters were tunneled subcutaneously to exit at the nape of the neck and guided through a metallic spring wire that was secured to the skin with suture material. The animals were allowed to recover, and then baseline values were obtained (cardiac output, mean arterial pressure, stroke volume, and base excess).

Thirty-two animals were used in the study. The sham burned (control) group consisted of eight animals that were subjected to the surgical interventions described above, were dipped in room-temperature water, and had hemodynamic parameters measured throughout the experiment. The 24 experimental animals were subjected to a 30% total body surface area (TBSA) full-thickness scald burn while the rats were under pentobarbital anesthesia. Rats were placed on a mold that allowed only the exposed skin to come into contact with 100°C water for 7 seconds. After the injury, the animals were immediately dried and placed in individual cages to recover from anesthesia. All rats had free access to water. The experimental animals were immediately resuscitated after the scald burn and followed for 6 hours. The resuscitation provided these animals was based on the Parkland formula (4 mL/kg for each 1% of TBSA burn), with 50% of the total calculated fluid requirement being allocated to the first 8 hours after burn. The fluid to be given per hour was calculated from this 8-hour volume; this volume per hour was given intravenously at a constant rate throughout the experiment, so that animals surviving 6 hours received three-quarters of the 8-hour ration. The rats were randomized into one of three treatment groups: group I ($n = 8$) received 4 mL/kg lactated Ringer's solution for each 1% of TBSA burn; group II ($n = 8$) received 2 mL/kg lactated Ringer's solution and 2 mL/kg human serum albumin for each 1% of TBSA burn; group III ($n = 8$) received 2 mL/kg lactated Ringer's solution and 2 mL/kg DCLHb for each 1% of TBSA burn. Table 1 offers a comparison of the different physiochemical properties of the resuscitative solutions.

For the remainder of the study, animals remained conscious and unrestrained and had indwelling venous and arterial catheters for access. At 2 hours after burn, all rats re-

TABLE 1. Physiochemical properties of diaspirin cross-linked hemoglobin, human serum albumin, and lactated Ringer's solution

	DCLHb (9.9 g/dL)	HAS (9.9 g/dL)	LR
Osmolality	300 mOsm/kg	300 mOsm/kg	273 mOsm/kg
pH	7.32	7.32	6.5
Na ⁺	140 mEq/L	140 mEq/L	130 mEq/L
K ⁺	3.9 mEq/L	3.9 mEq/L	4 mEq/L
Ca ²⁺	2.0 mEq/L	2.0 mEq/L	2.7 mEq/L
Mg ²⁺	0.9 mEq/L	0.9 mEq/L	
Cl ⁻	120 mEq/L	120 mEq/L	109 mEq/L
Lactate	34 mEq/L	34 mEq/L	28 mEq/L

HAS, human serum albumin; LR, lactated Ringer's solution.

ceived a dose of buprenorphine (0.3 mg/kg) intravenously for pain control. Any animal that showed signs of pain or discomfort received an additional dose of buprenorphine intravenously as needed. The animals that survived to 6 hours were killed.

The femoral artery was used to monitor mean arterial pressure (Columbus Instruments). Cardiac output and stroke volume were measured by thermodilution (Columbus Instruments). Systemic vascular resistance was calculated as mean arterial pressure/cardiac output. Arterial blood gases were obtained through the femoral artery catheter and analyzed for base excess (BE) using a Corning 168 Ph blood gas analyzer (Corning Glassworks, Medfield, Mass). BE was determined by the microanalyzer based on the following equation: $BE_{\text{extracellular fluid}} = [HCO_3^-] - 25 + 16.2 (\text{pH } 7.40)$. Values were obtained at baseline (after line placements but before burn) and at 1, 2, 3, 4, 5, and 6 hours after burn. The volume of blood taken as a sample from each animal was 1 mL/h, including the baseline value, for a total of 7 mL of blood for the entire experiment. The blood drawn was not replaced in any of the groups, including the control group.

The DCLHb was prepared by Baxter Healthcare Corp. as described previously.¹¹ The DCLHb was formulated as 10 g/dL hemoglobin in a balanced electrolyte solution (<10% methemoglobin, pH 7.4, at 37°C; P50 = 32 mm Hg; oncotic pressure = 42 mm Hg). Before infusion, the sterile solution was filtered through a 0.2- μ m cellulose acetate filter. The buffered human serum albumin (Baxter Healthcare) was oncologically matched to 10% DCLHb.

Data are expressed as means $\pm$ SEM. Statistical analysis was performed using two-way analysis of variance followed by multiple comparisons with the Dunnett test or the Turkey-Kramer test, where appropriate, to determine the degree of statistical significance of differences within and between groups. The Fisher's exact test was used to test for differences in survival between groups. Results were considered to be significant at $p < 0.05$. The number of animals used in each group ($n = 8$) was the minimum number required to produce statistically significant differences in mortality rates.

RESULTS

Mean Arterial Pressure

There were no significant differences in mean arterial pressure, cardiac output, stroke volume, systemic vascular resistance, and base excess between any of the groups at baseline.

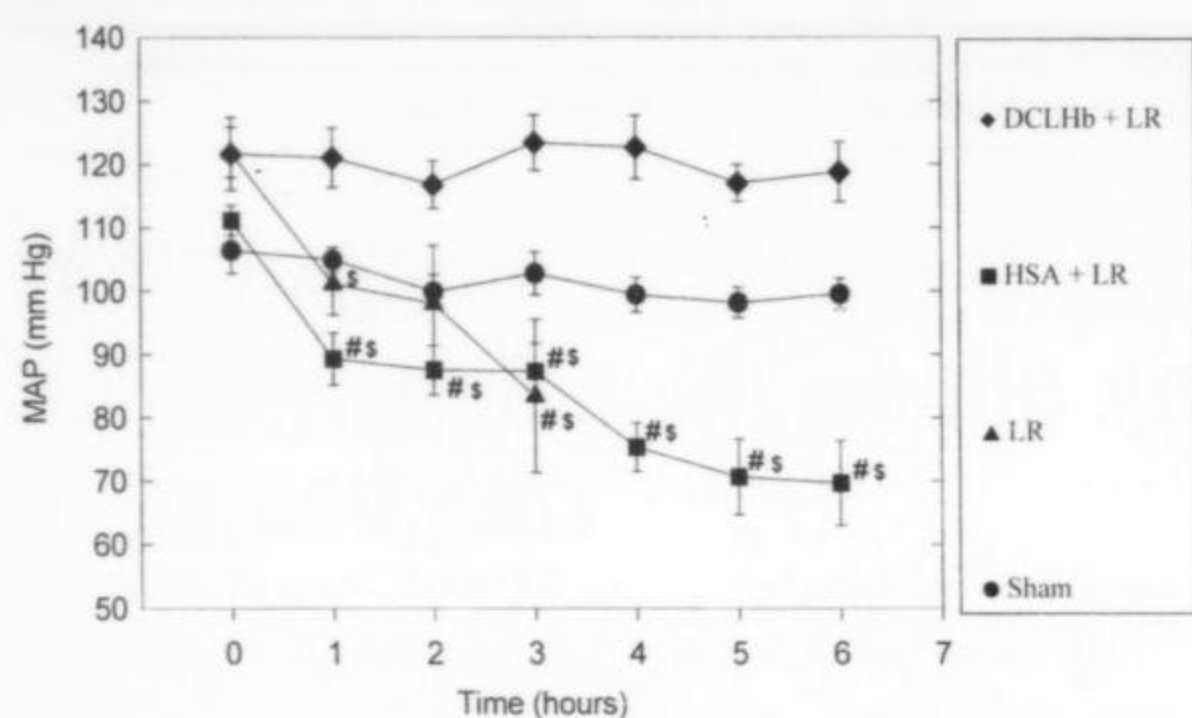


FIG 1. Changes in mean arterial pressure (MAP; mm Hg) noted after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (#) $p < 0.05$ compared with DCLHb-lactated Ringer's solution group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

One hour after thermal injury, there was a decrease in cardiac output, stroke volume, and base excess and an increase in systemic vascular resistance in all treatment groups, indicating development of the hypodynamic stage of burn shock. In rats resuscitated with DCLHb-lactated Ringer's solution, mean arterial pressure was maintained within baseline values throughout the study (Fig. 1). This was not the case with the animals that received either lactated Ringer's solution alone or human serum albumin-lactated Ringer's solution; these two groups showed a significant decrease in mean arterial pressure ($p < 0.05$) throughout the study compared with either their baseline value or the DCLHb-lactated Ringer's solution group at the same time point (Fig. 1).

Base Deficit

At baseline, all animals had a moderate base deficit, probably secondary to surgical trauma and anesthesia (Fig. 2). One

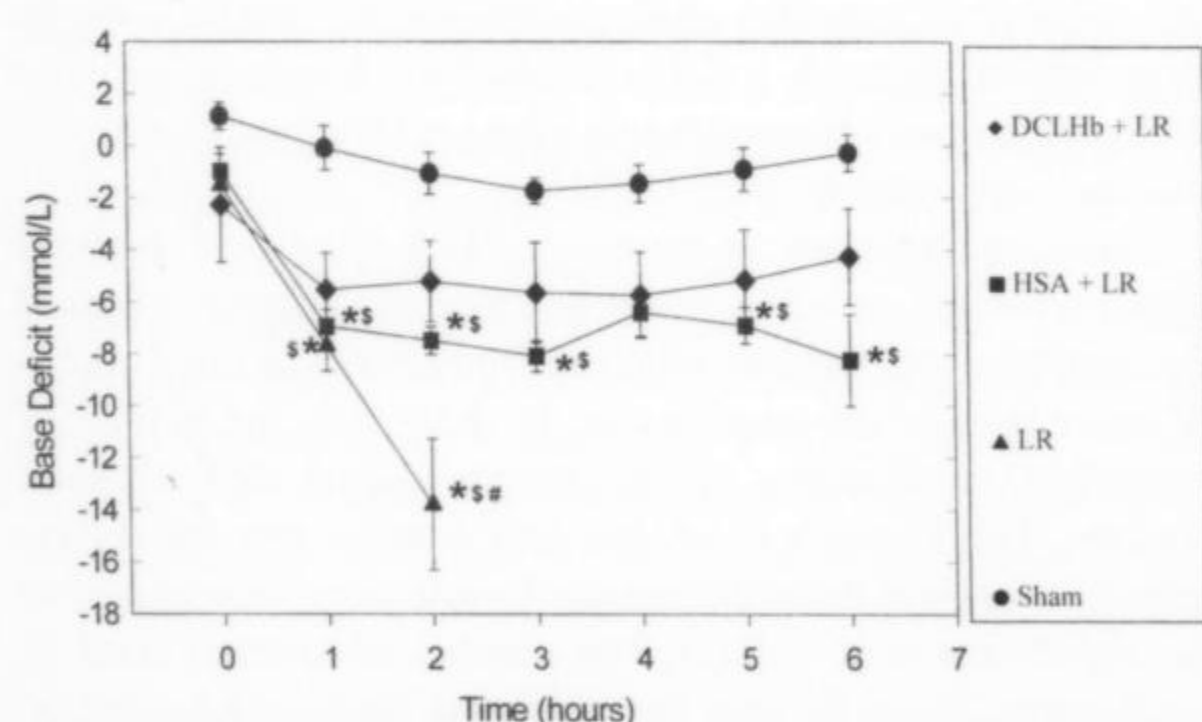


FIG 2. Changes in base deficit (mmol/L) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group; (#) $p < 0.05$ compared with the DCLHb-lactated Ringer's solution group at the same time point. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

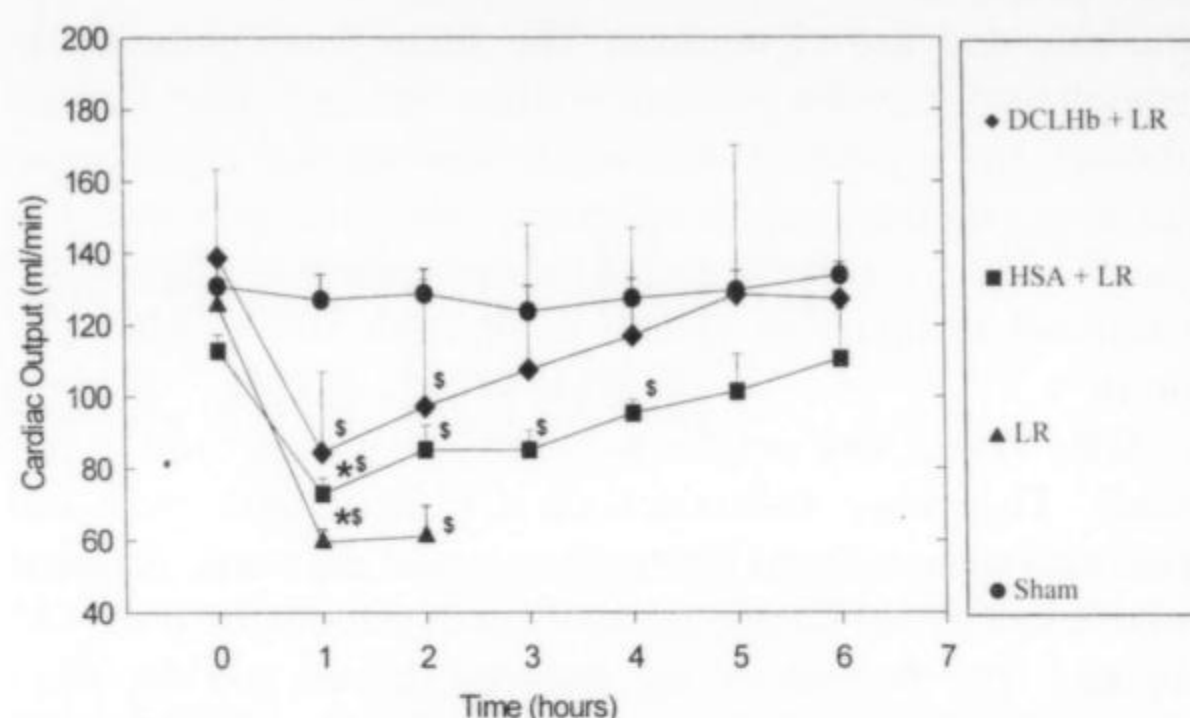


FIG 3. Changes in cardiac output (mL/min) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

hour after thermal injury, there was a worsening in base deficit in all treatment groups. The rats resuscitated with lactated Ringer's solution became increasingly acidotic, with an average base deficit of -14.0 mmol/L at 2 hours after burn. These results were extremely significant ($p < 0.01$) compared with the baseline value within the group or with the DCLHb-lactated Ringer's solution group and the sham group at the same time point (Fig. 2). The animals that were infused with human serum albumin-lactated Ringer's solution also experienced a significant ($p < 0.05$) progressive acidosis with an increasing base deficit compared with baseline values and with the sham group at the same time point; however, their values did not decrease as far as those of the group that was resuscitated with only lactated Ringer's solution. In contrast, there was no significant difference in base deficit between the sham group and the DCLHb-lactated Ringer's solution group throughout the study (Fig. 2).

Cardiac Output and Stroke Volume

There was a significant decrease in cardiac output and stroke volume in all treatment groups ($p < 0.05$) at 1 hour after burn compared with baseline values within the groups (Figs. 3 and 4). The animals resuscitated with lactated Ringer's solution alone remained hypodynamic throughout the study, and both the cardiac output and the stroke volume failed to return to baseline values. The rats resuscitated with human serum albumin-lactated Ringer's solution also had decreased cardiac output and stroke volume compared with baseline values within the group ($p < 0.05$); however, by 5 and 6 hours, respectively, their cardiac output and stroke volume returned to baseline values. In contrast, the administration of DCLHb-lactated Ringer's solution returned cardiac output and stroke volume to baseline levels at 3 and 2 hours after burn, respectively (Figs. 3 and 4).

Systemic Vascular Resistance

There was a significant increase in systemic vascular resistance in the DCLHb-lactated Ringer's solution group and the

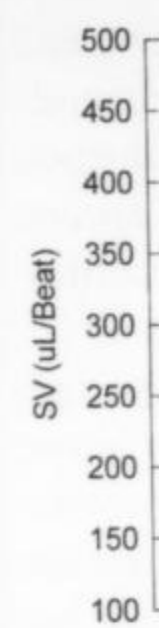


FIG 4. Changes in stroke volume (uL/Beat) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

lactated Ringer's solution group ($p < 0.05$) and with the sham group at the same time point. The animals that were infused with human serum albumin-lactated Ringer's solution also experienced a significant ($p < 0.05$) progressive acidosis with an increasing base deficit compared with baseline values and with the sham group at the same time point; however, their values did not decrease as far as those of the group that was resuscitated with only lactated Ringer's solution. In contrast, there was no significant difference in base deficit between the sham group and the DCLHb-lactated Ringer's solution group throughout the study (Fig. 2).

There was a significant decrease in cardiac output and stroke volume in all treatment groups ($p < 0.05$) at 1 hour after burn compared with baseline values within the groups (Figs. 3 and 4). The animals resuscitated with lactated Ringer's solution alone remained hypodynamic throughout the study, and both the cardiac output and the stroke volume failed to return to baseline values. The rats resuscitated with human serum albumin-lactated Ringer's solution also had decreased cardiac output and stroke volume compared with baseline values within the group ($p < 0.05$); however, by 5 and 6 hours, respectively, their cardiac output and stroke volume returned to baseline values. In contrast, the administration of DCLHb-lactated Ringer's solution returned cardiac output and stroke volume to baseline levels at 3 and 2 hours after burn, respectively (Figs. 3 and 4).

FIG 5. Changes in systemic vascular resistance (mmHg/L/min) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group; (#) $p < 0.05$ compared with the DCLHb-lactated Ringer's solution group at the same time point. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

TABLE

Sham group
a $p < 0.05$
b $p < 0.05$

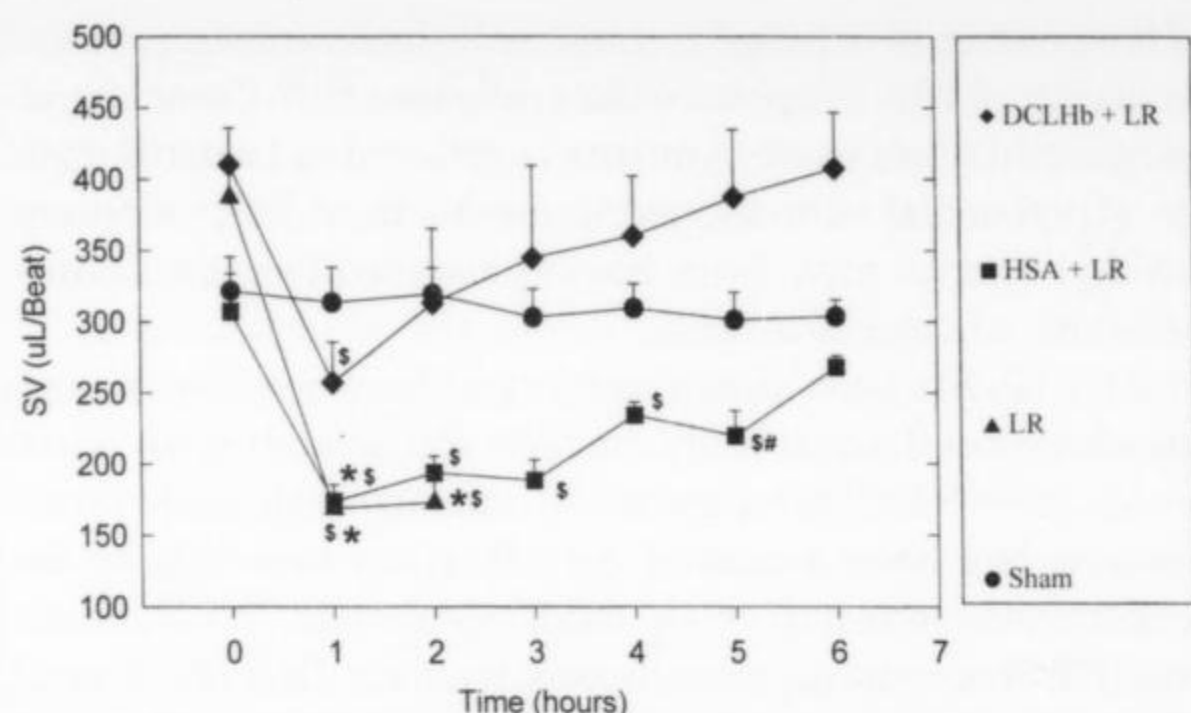


FIG 4. Changes in stroke volume (SV; μL per beat) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group; (#) $p < 0.05$ compared with the DCLHb-lactated Ringer's solution group at the same time point. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

lactated Ringer's solution only group at 1 hour after burn ($p < 0.05$) compared with baseline values within the groups and with the sham group at the same time point (Fig. 5). The animals resuscitated with lactated Ringer's solution alone remained hypodynamic throughout the study, and the systemic vascular resistance failed to return to baseline values.

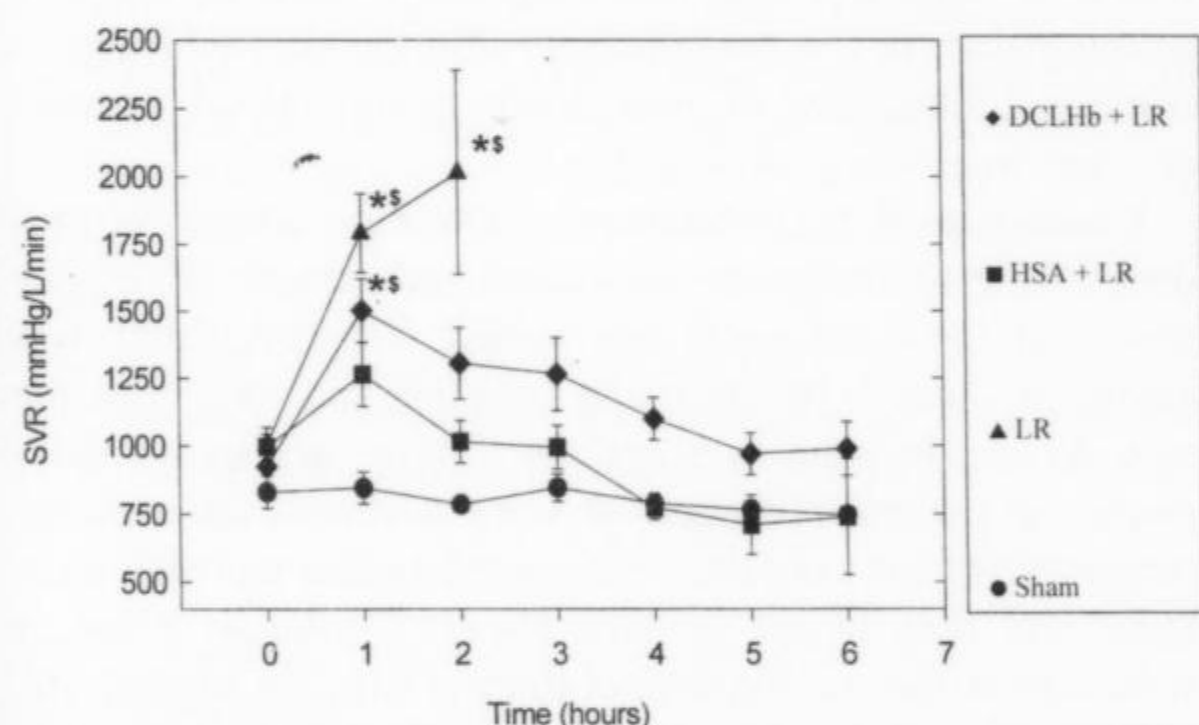


FIG 5. Changes in systemic vascular resistance (SVR; $\text{mm Hg} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$) after burn injury. There were eight animals per treatment group. If the number of surviving animals decreased to less than three or if no hemodynamic data could be obtained from the remaining surviving rats, the line was terminated at that time point. (*) $p < 0.05$ compared with the sham group at the same time point; (\$) $p < 0.05$ compared with the baseline value within the group. Values are means $\pm$ SEM. Sham, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

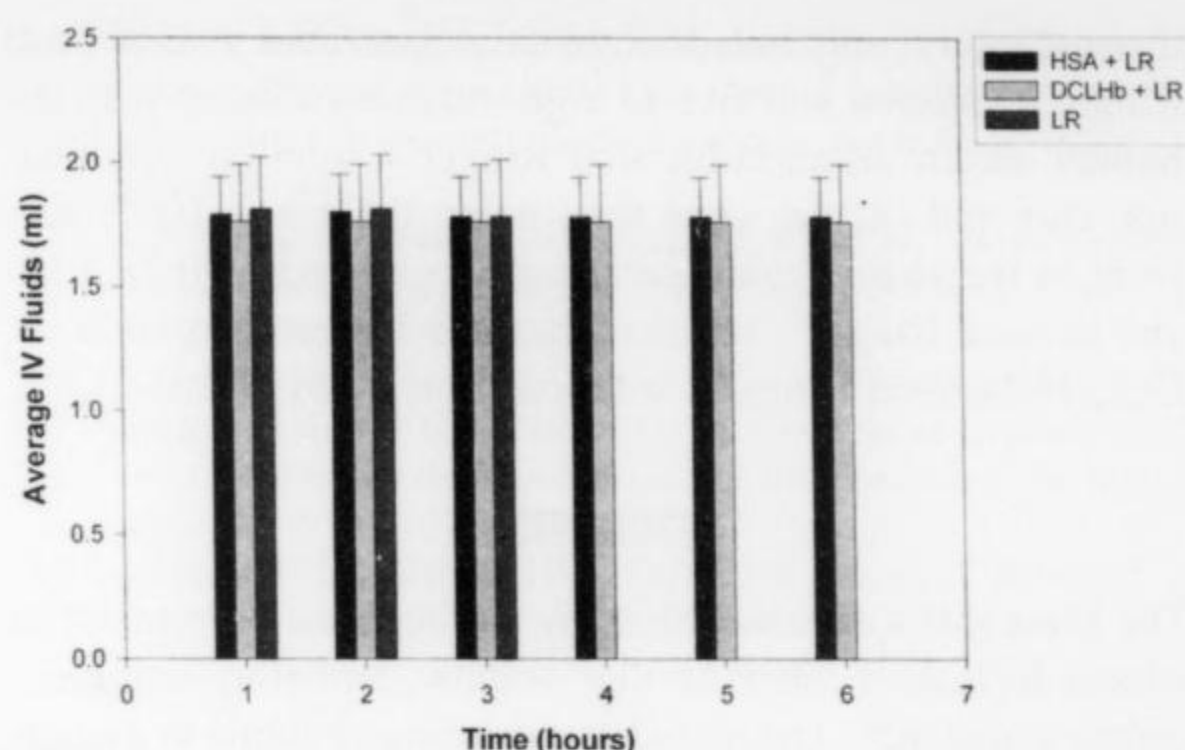


FIG 6. Intravenous fluid (mL) administered per hour after burn injury. Resuscitation was based on the Parkland formula (4 mL/kg for each 1% of total body surface area burn), with half of the total fluid administered during the first 8 hours. The fluid to be given per hour was calculated from this 8-hour volume; this volume per hour was given intravenously at a constant rate throughout the experiment, so that animals surviving 6 hours received three-quarters of the 8-hour ration. Animals were followed for 6 hours. There were eight animals per treatment group. If the number of surviving animals decreased to less than three, the bar graph was terminated at that time point. LR, lactated Ringer's solution, HSA, human serum albumin.

In contrast, by 2 hours after burn and for the remainder of the study there was no significant difference in systemic vascular resistance between the DCLHb-lactated Ringer's solution group and the sham group at the same time points (Fig. 5). The rats resuscitated with human serum albumin-lactated Ringer's solution suffered an increase in systemic vascular resistance, but these values were not significant compared with the sham group or with the baseline values (Fig. 5).

Intravenous Volume Administered

As discussed above, the resuscitation was based on the Parkland formula. There was no statistically significant difference between the experimental groups in the amount of intravenous volume administered throughout the resuscitation period (Fig. 6).

Mortality

Three of eight animals resuscitated with lactated Ringer's solution and six of eight animals treated with human serum albumin-lactated Ringer's solution survived after 4 hours of thermal injury, whereas all rats resuscitated with the DCLHb-lactated Ringer's solution combination survived the first 4 hours after burn (lactated Ringer's solution vs. DCLHb-lactated Ringer's solution; $p < 0.05$) (Table 2). At 6 hours after

TABLE 2. Cumulative mortality rates at different time points in 32 rats in this burn shock model

Treatment Group	Time after Intravenous Resuscitation (hr)					
	1	2	3	4	5	6
Sham	0 of 8	0 of 8	0 of 8	0 of 8	0 of 8	0 of 8
DCLHb + LR	0 of 8	0 of 8	0 of 8	0 of 8	0 of 8	0 of 8
HSA + LR	1 of 8	2 of 8	2 of 8	2 of 8	2 of 8	3 of 8
LR	0 of 8	2 of 8	4 of 8	5 of 8 ^{a,b}	5 of 8 ^{a,b}	6 of 8 ^{a,b}

Sham, control group, not burned, not resuscitated; LR, lactated Ringer's solution; HSA, human serum albumin.

^a $p < 0.05$ compared with the DCLHb + LR group.

^b $p < 0.05$ compared with the sham group.

thermal injury, only two of eight animals treated with lactated Ringer's solution and five of eight rats resuscitated with the human serum albumin-lactated Ringer's solution combination survived. At the same time point, there was 100% survival in the sham group and the group treated with DCLHb and lactated Ringer's solution (lactated Ringer's solution vs. DCLHb-lactated Ringer's solution; $p < 0.05$) (Table 2).

DISCUSSION

The fluid shifts after a 30% body surface area burn result in edema formation, intravascular volume depletion, and electrolyte imbalance. Decreased intravascular volume is associated with changes in cardiovascular function, which include reduced cardiac output and increased systemic vascular resistance. It is now known, however, that intense vasoconstriction and decreased cardiac performance persist in spite of massive fluid resuscitation.¹² These factors present management dilemmas to the clinician, because persistent tissue and organ ischemia in the face of massive fluid administration frequently result in increased tissue edema and eventual volume overload. As our knowledge of these various changes increases, different means of blocking or augmenting the burn shock physiologic responses may become available. Future therapies might permit resuscitation with administration of less fluid and better organ function.

The host's inflammatory response to thermal injury plays a key role in the development of burn shock.¹³⁻¹⁵ Systemic host defenses are activated, including plasma factors (e.g., complement and clotting cascades) and cellular components (e.g., neutrophils, monocytes, macrophages, and endothelial cells). These activated cells release endogenous inflammatory mediators, including nitric oxide, which augment the inflammatory response. Hypovolemia is another common factor seen in unresuscitated thermal injury. If left unchecked, the combination of these events can lead to shock, systemic inflammatory response syndrome, and eventually death.

Nitric oxide has been identified as an important mediator of the hypotension associated with septic shock.¹⁶ Overproduction of nitric oxide can lead to direct tissue injury and organ failure.¹⁷ The synthesis of nitric oxide is regulated by two different enzymes. In endothelial cells, the constitutive calcium-dependent nitric oxide synthase enzyme is responsible for the basal release of nitric oxide that maintains normal vasodilatory vascular tone and participates in intracellular communication.¹⁸⁻²⁰ The second type of nitric oxide synthetase enzyme is calcium-independent and is induced by inflammatory mediators in endothelial cells and vascular smooth muscle.^{21,22} This inducible form of nitric oxide synthase is not controlled by a receptor-dependent mechanism and is believed to be responsible for the high plasma levels of nitric oxide in septic patients.²³ Recently, high levels of nitric oxide have been found in burn patients.²⁴ We believe that this free radical could be partially responsible for the hypodynamic state seen in the early stages of burn shock.

In animal models, the effects of nitric oxide inhibition on hemodynamics, gas exchange, survivability, and oxygen transport have been studied using L-arginine methyl ester.

This product is believed to bind to both the constitutive and inducible forms of nitric oxide synthetase.^{25,26} Complete inhibition of nitric oxide synthesis is believed to be detrimental to experimental animals, and in anesthetized rats, L-arginine methyl ester in high doses has been shown to cause cardiovascular collapse and death.²⁷

DCLHb has been shown to elevate blood pressure through mechanisms that probably include the inhibition of nitric oxide.^{10,28,29} DCLHb appears to interact with nitric oxide after it has been produced by either the inducible or the constitutive forms of nitric oxide synthetase.^{10,28,29} In our study of burn shock, resuscitation with the DCLHb-lactated Ringer's solution combination prevented hypotension and improved cardiac output, stroke volume, and survivability. DCLHb appears to improve hemodynamics by redistributing blood flow from musculoskeletal tissues rather than from critical organs.³⁰ The beneficial effects of DCLHb could also be secondary to the inhibition of nitric oxide. This hypothesis is supported by the fact that in the DCLHb-treated group we see improved hemodynamic parameters combined with a modest increase in systemic vascular resistance. It has been shown that DCLHb is superior to lactated Ringer's solution, human serum albumin, and blood in restoring hemodynamic stability, base excess, central venous oxygen saturation, and subcutaneous and intestinal tissue oxygenation in various models of shock.⁶⁻⁸ In this model of thermal injury, our results demonstrate that the DCLHb-lactated Ringer's solution combination proved to be superior to lactated Ringer's solution alone or to a human serum albumin-lactated Ringer's solution combination in normalizing hemodynamic parameters and improving survivability.

Under normal circumstances, a 30% burn would not produce a 75% mortality rate in resuscitated animals. To clarify this point, the sham group (not burned, not resuscitated) was studied to observe the effects of sampling and surgery on the rats. As can be seen in all of the figures, neither hemodynamic parameters nor survival were adversely affected by surgery or blood sampling. We speculate that the high mortality rate seen in the lactated Ringer's solution group is secondary to the combination of three events: the surgery, the 30% burn, and the blood sampling. This fluid collection probably added a hemorrhagic shock component to the model. The animals that received DCLHb-lactated Ringer's solution, however, successfully tolerated these three insults. The question that arises then is, what is the effect of hemodilution on these animals? And would fresh blood work as well or better than the DCLHb in this model? To answer these questions, we are now conducting a study comparing fresh blood with DCLHb in this burn shock model.

The primary purpose of this project was to evaluate and compare the current standard of care (the Parkland formula) and a modified regimen incorporating DCLHb; however, further studies are needed to define the optimal dose of DCLHb. Previous experiments have looked at the efficacy of colloid versus crystalloid resuscitation after thermal injury. Colloid has been shown to improve hemodynamics in the early stages of burn shock; however, a clinical trial performed at Brooke Army Medical Center reported that the use of

colloid linked DCLHb possibly reported

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REFER

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colloid resuscitation in the early stages of burn shock was linked with increased pulmonary edema.³¹ If low doses of DCLHb are used for resuscitation in burn shock, it may be possible to avoid the detrimental side effects previously reported with colloid resuscitation.

In summary, in this model of burn shock, the DCLHb-lactated Ringer's solution combination was superior to lactated Ringer's solution alone and the human serum albumin-lactated Ringer's solution combination in maintaining mean arterial pressure and improving cardiac output, stroke volume, base excess, and survival for up to 6 hours after thermal injury. These findings suggest that administration of DCLHb may be useful in the resuscitation scheme in burn shock.

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